RULES OF BONDING

General Chemistry I, Lecture Series 10

Qixi Mi

Reading:

OGB8 §3.10, §3.11

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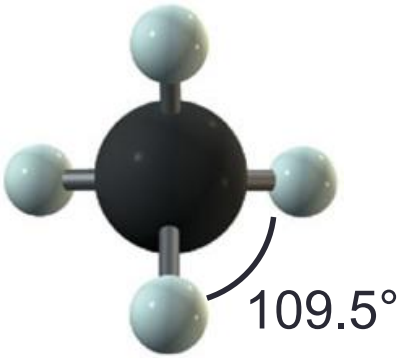
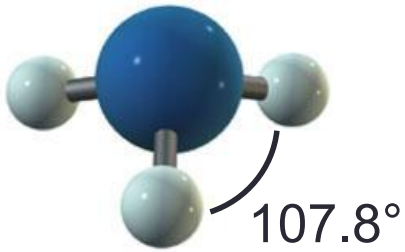
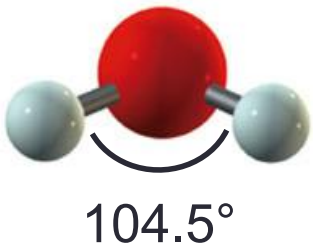
Polyatomic Molecules



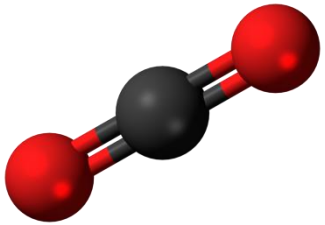
CH_4



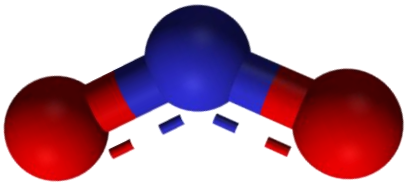
NH_3



H_2O



CO_2



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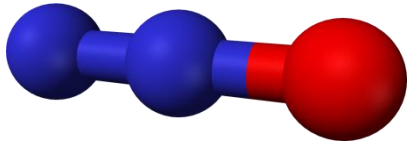
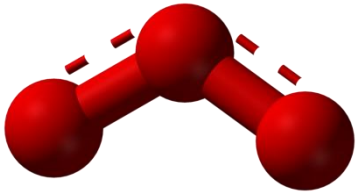


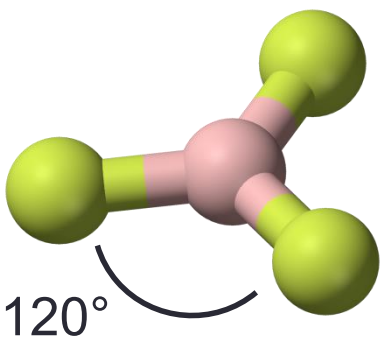
N_2O



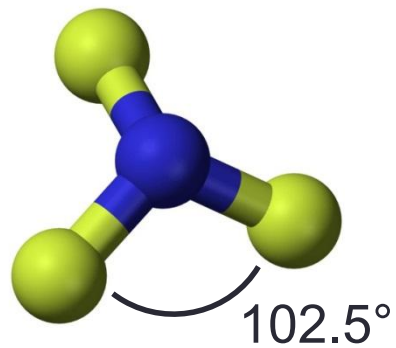
O_3

Polyatomic Molecules: Bond Angles

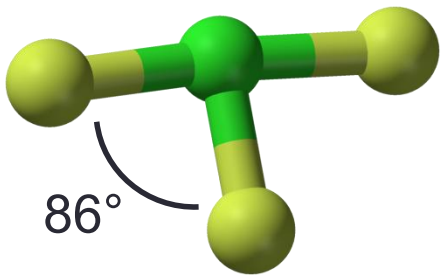
					He
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr



BF_3

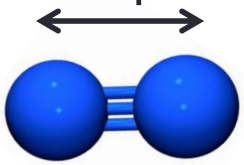


NF_3

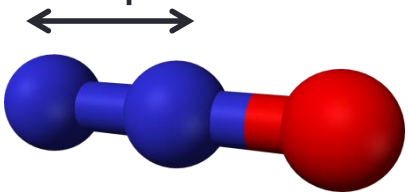


BrF_3

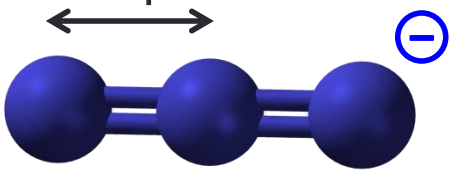
Polyatomic Molecules: Bond Lengths



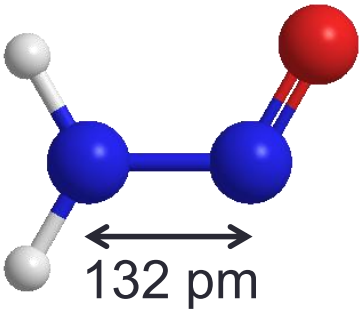
N_2



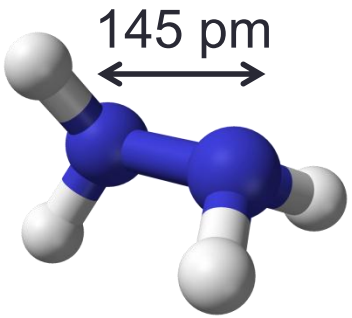
N_2O



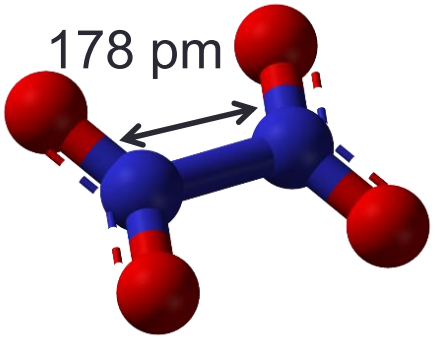
N_3^-
Azide 叠氮



H_2NNO
Nitrosamine 亚硝胺



N_2H_4
Hydrazine 肼

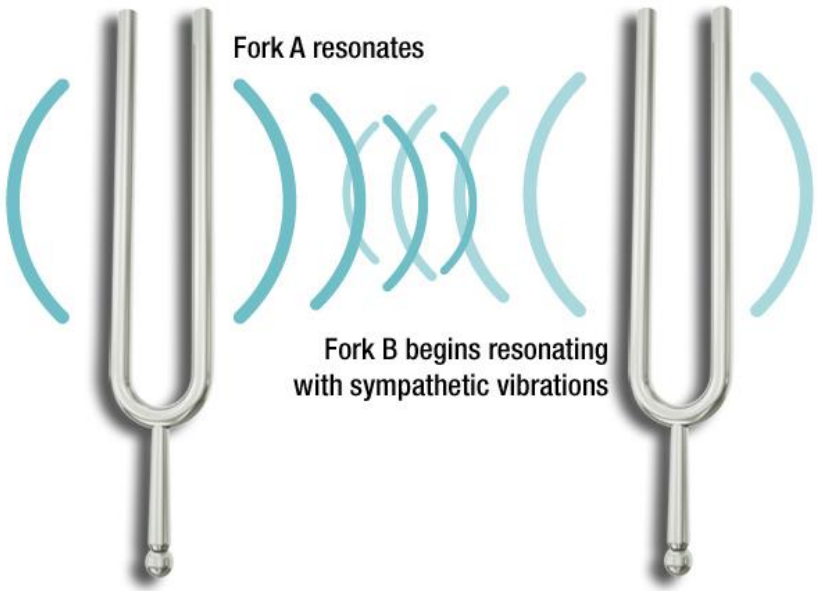


N_2O_4

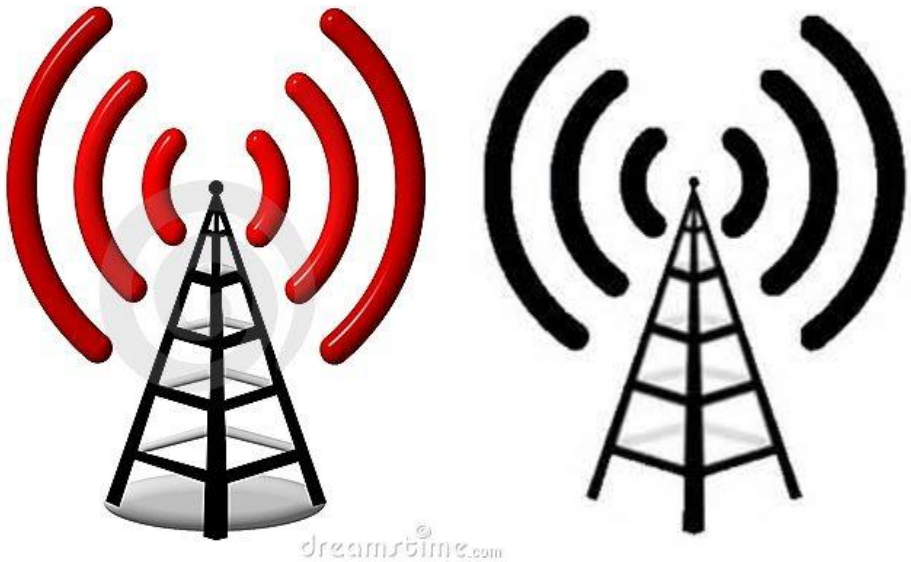
Outline

- Lewis dot structures (1916)
- Resonance structures (1930s)
- VSEPR (1958)

What is Resonance?



Acoustic resonance



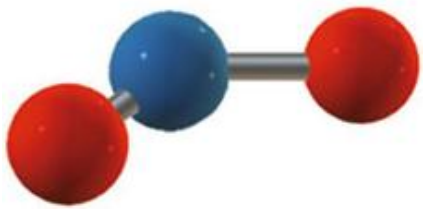
Electromagnetic resonance

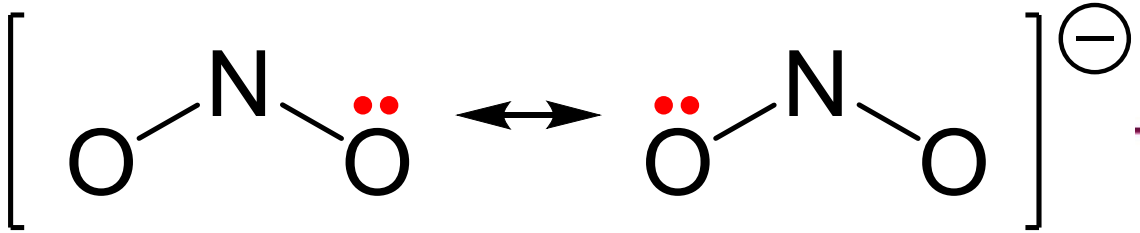
1. Two separate units
2. Identical or very close *E* or *v*
3. Moderate coupling 耦合

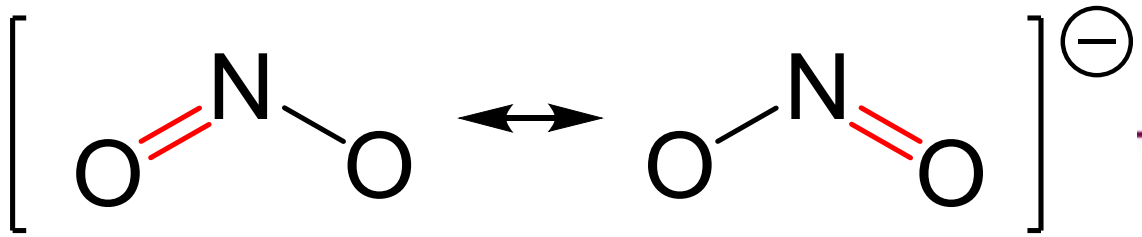
Resonance in Chemistry

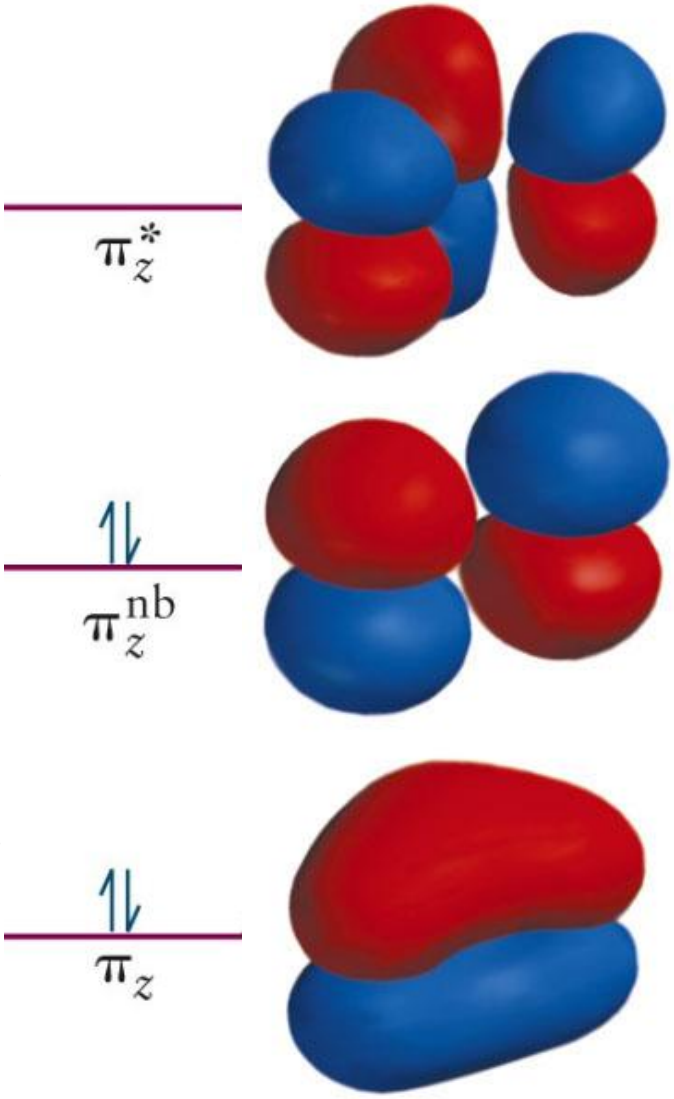
OGB8 p.273

NO_2^-

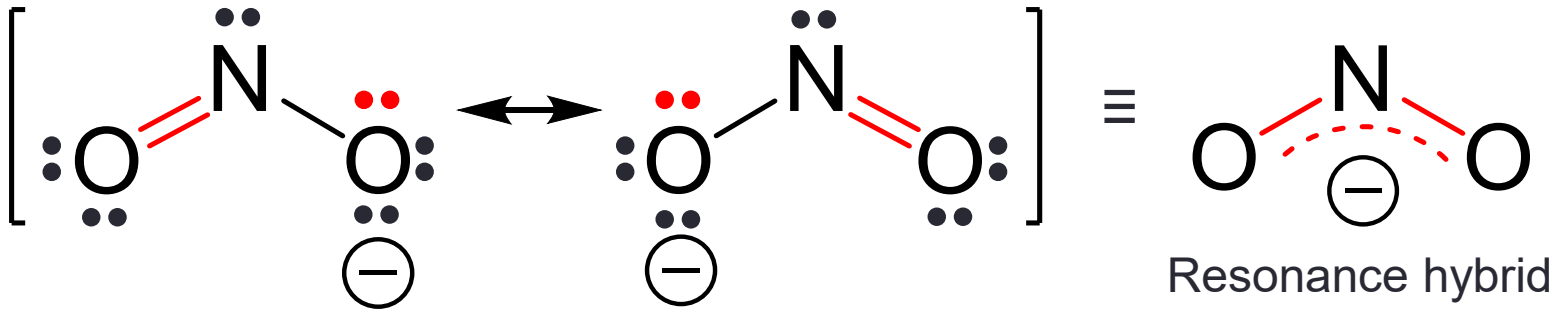






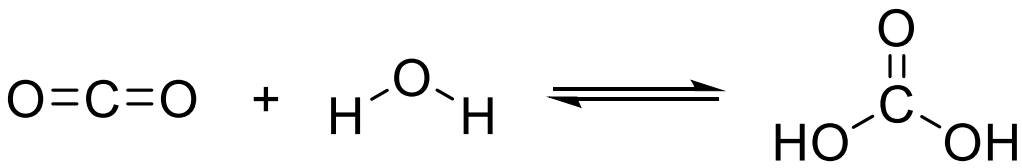


Resonance Structures

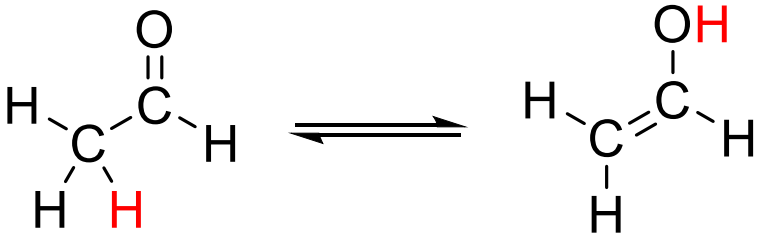


Resonance hybrid

1. Canonical forms 正则形式
2. Identical or close *E*
3. Delocalized bonding 离域键



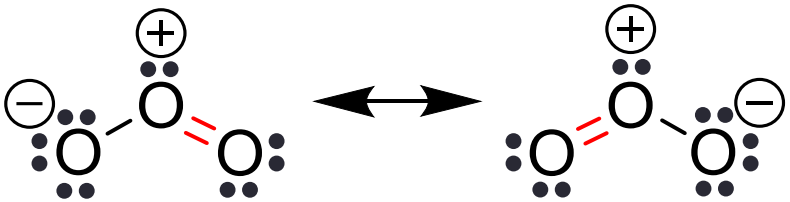
Chemical equilibrium
化学平衡



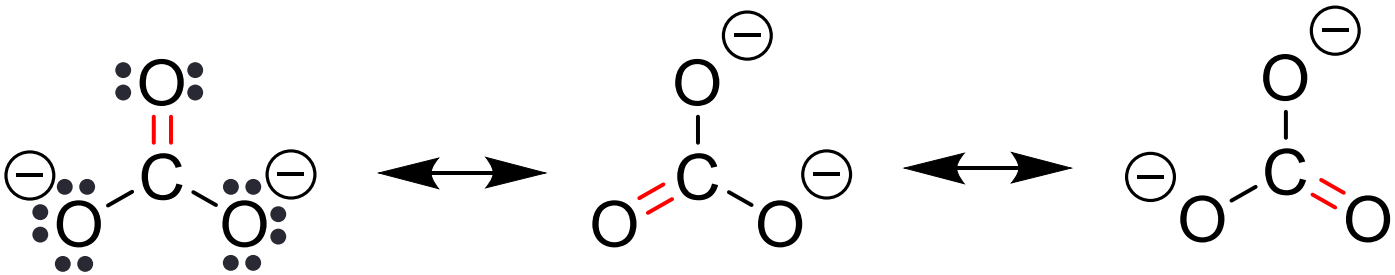
Tautomerism
互变异构

Resonance Structure and Bond Order

$$\text{B.O.(Resonance hybrid)} \approx \overline{\text{B.O.(Canonical forms)}}$$

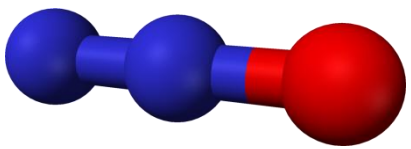


O–O bond order = 1.5

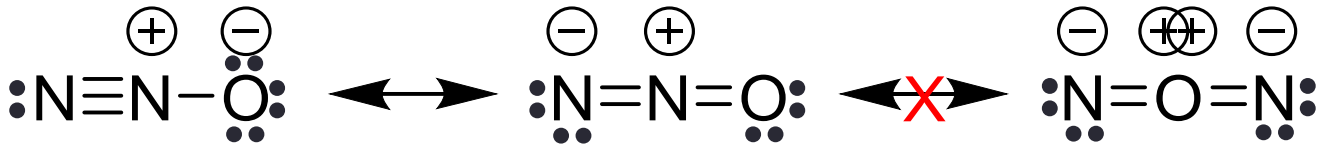


C–O bond order ≈ 1.3

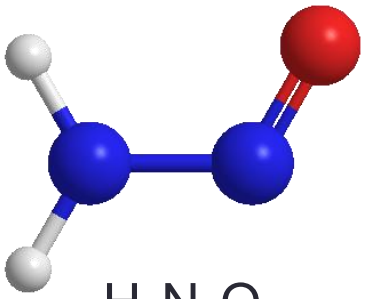
Asymmetric Resonance Structures



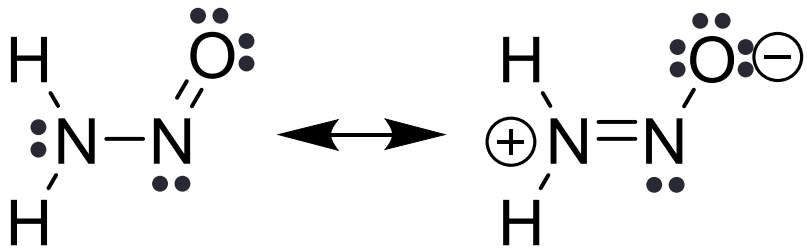
N_2O



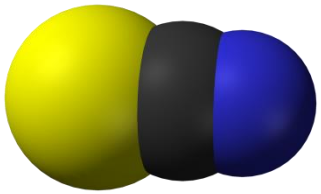
N–N bond order ≈ 2.5



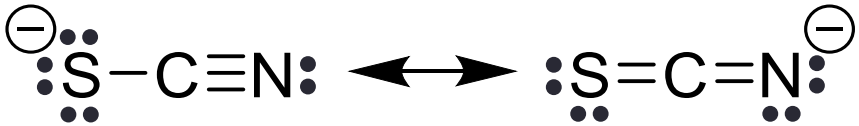
H_2N_2O



N–N bond order ≈ 1.5



SCN^-
thiocyanate
硫氰酸根



C–N bond order ≈ 2.5

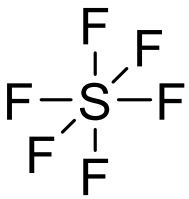
Number of Valence Electrons

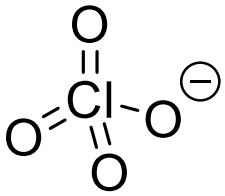
Duet

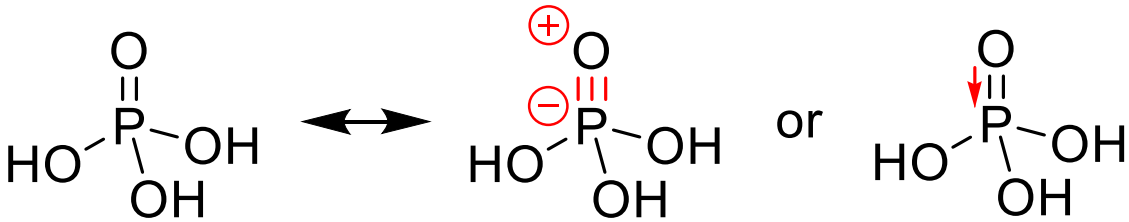
6, 8 Octet 八隅体

1*s*	H							He
2*s*, 2*p*	Li	Be	B	C	N	O	F	Ne
3*s*, 3*p*, 3*d*	Na	Mg	Al	Si	P	S	Cl	Ar
	K	Ca	Ga	Ge	As	Se	Br	Kr
	Rb	Sr	In	Sn	Sb	Te	I	Xe
	Cs	Ba	Tl	Pb	Bi	Po	At	Rn
	Fr	Ra						

8, 10, **12**, 14, …







2 < BO(P=O) < 3

Back coordinate bond
反馈配位键

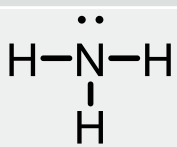
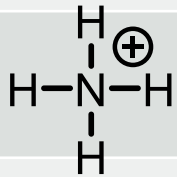
Valence *vs*. Oxidation Number

Valence 成键数

= Number of shared electrons

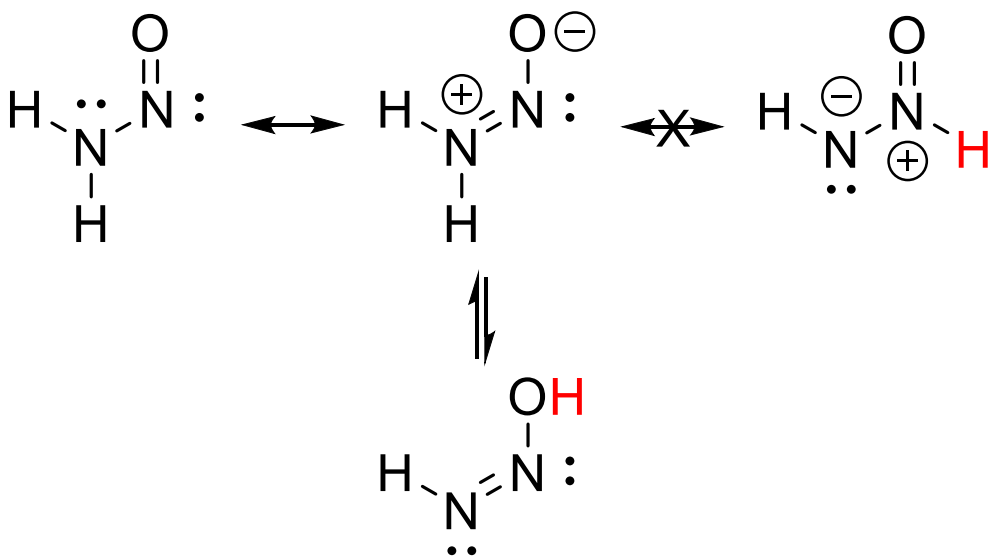
Oxidation number 氧化数

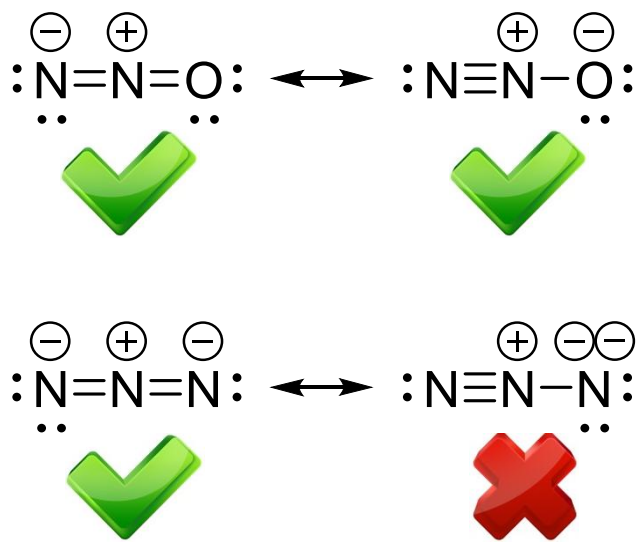
= Number of lost electrons

F^-, Cl^-	V = 0	ON = −1	V < |ON|
	V = 3	ON = −3	V = |ON|
	V = 4	ON = −3	V > |ON|
Cl−Hg−Hg-Cl	V = 2	ON = 1	V > |ON|

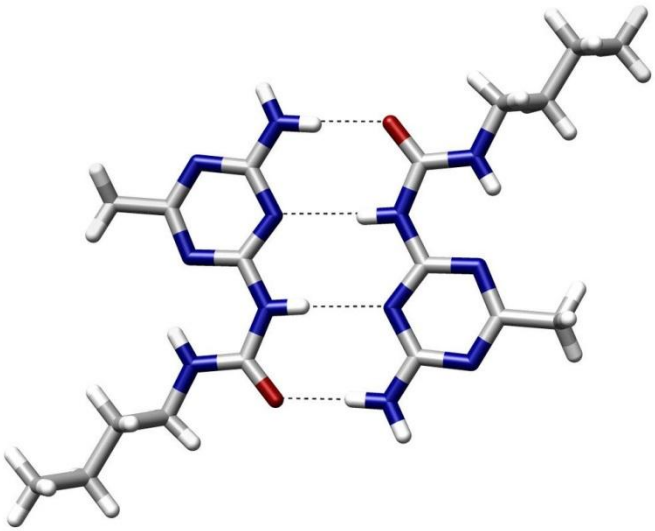
Rules of Resonance Structures

- Valid Lewis dot structure
- Fixed atomic positions
- Lowest energy
  - Follow the trend of electronegativity
  - The more bonds, the more stable
  - More resonance structures → Better delocalization



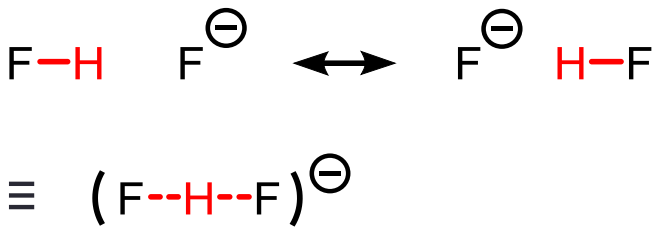


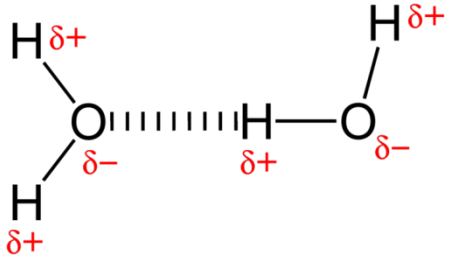
The Hydrogen Bond

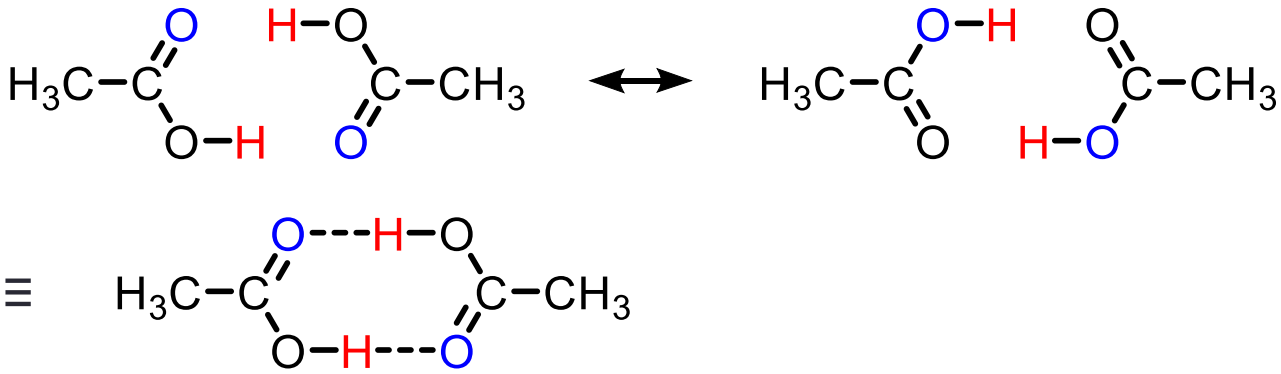


- Attractive force between X−H and Y (X, Y = F, O, N)

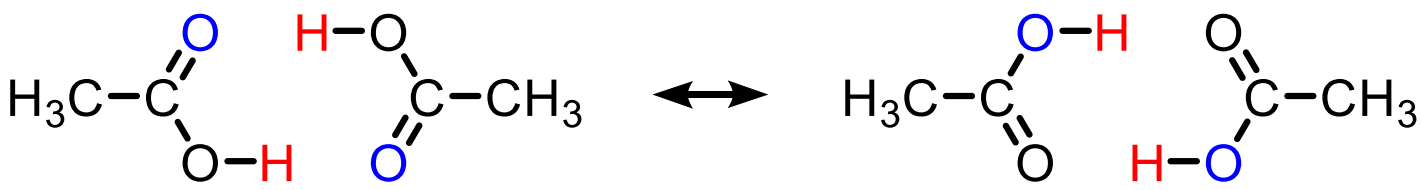
- H is unique being light and mobile







Summary



Electron

Hydrogen

Heavy atoms
(C, N, O, …)

Survey

Which of the following statements about resonance structures is incorrect?

A. Resonance structures normally have fixed atomic positions.

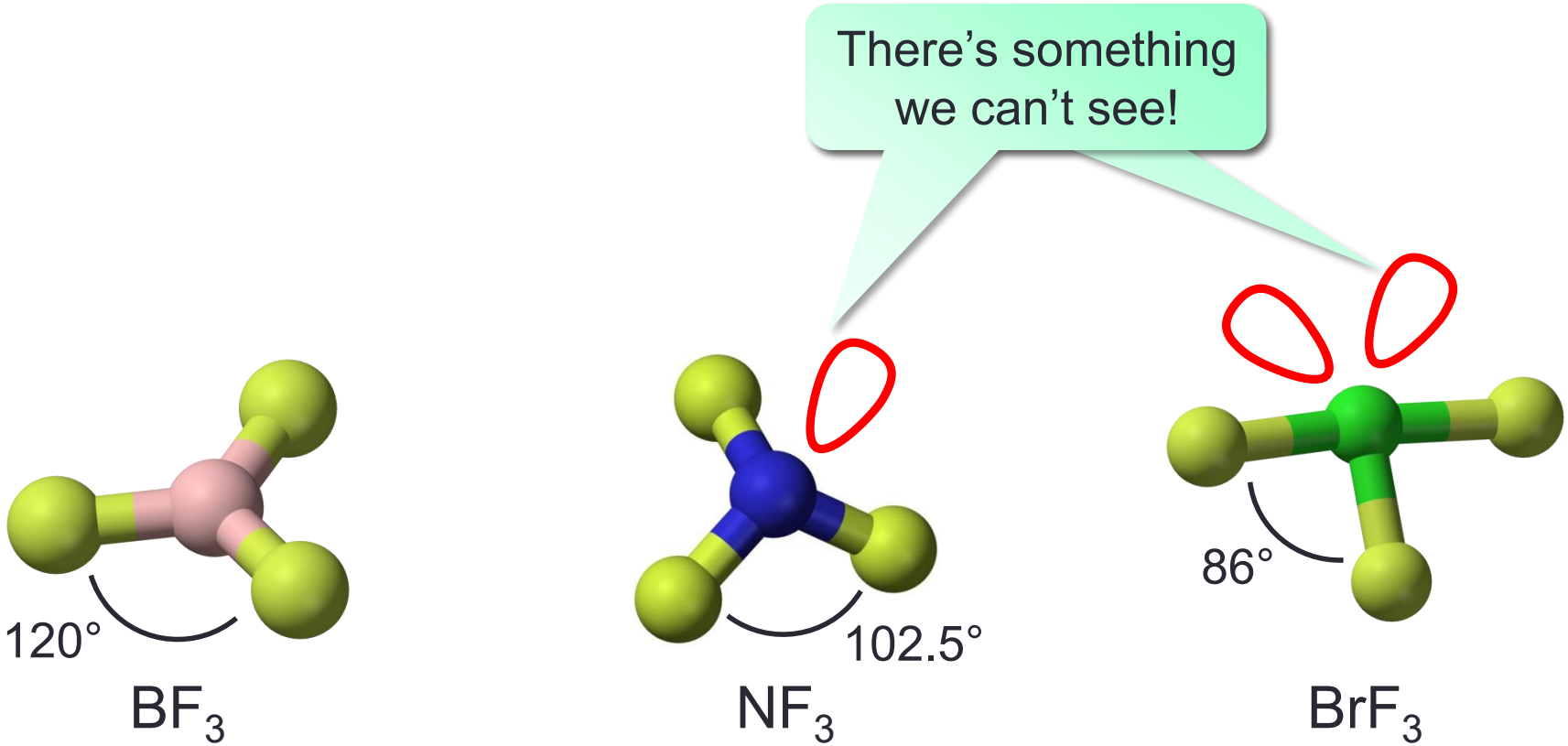
B. Resonance structures normally have the same total number of covalent bonds.

C. Resonance structures normally have the same formal charges.

Outline

- Lewis dot structures (1916)
- Resonance structures (1930s)
- VSEPR (1958)

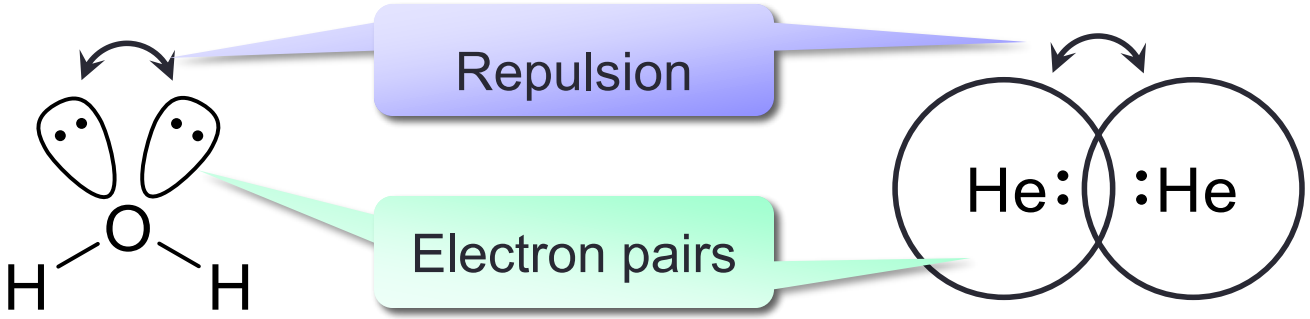
Molecular Geometry



The Valence Shell Electron-Pair Repulsion Theory

Ideas based on VB theory:

- Only the valence shell matters
- Shared electron pairs and lone pairs
- Molecular energy ≈
  Bond energies + repulsion energies



Ronald J. Gillespie
(UCL, 1924–2021)

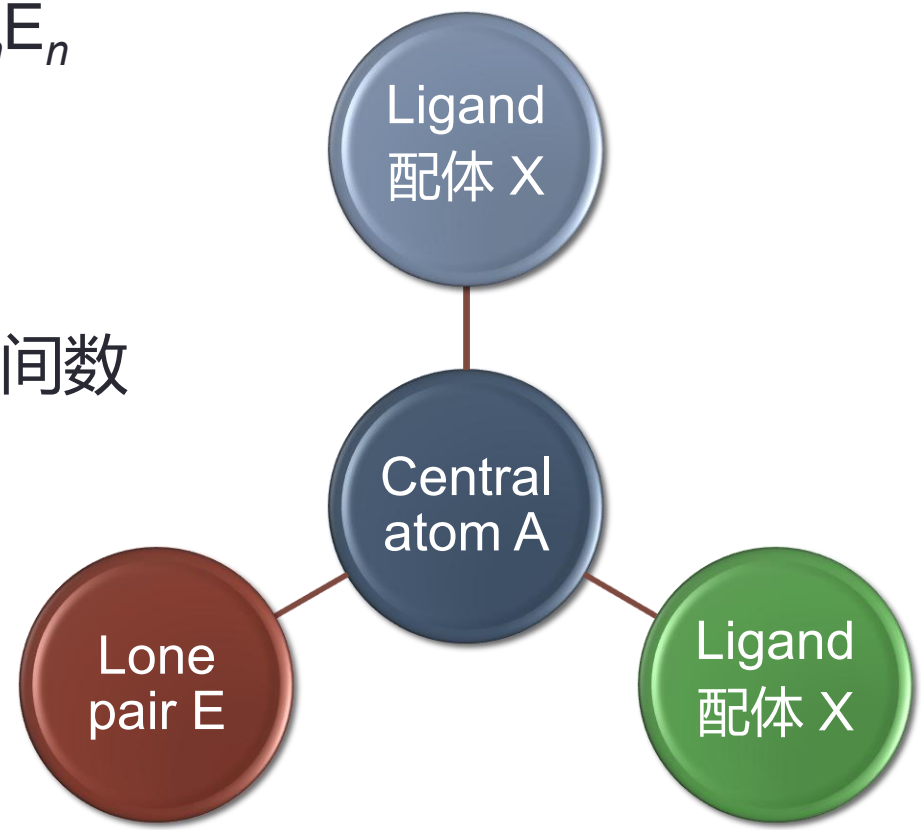
Ronald S. Nyholm
(UCL, 1917–1971)

Rules of VSEPR Theory

1. Rewrite a structure as AX_mE_n
   m = number of ligands
   n = number of lone pairs

2. Assign a Steric Number 空间数
   SN = $m + n$

3. Place the ligands and lone pairs on a polyhedron of SN vertexes



4. Size: Lone pair > multiple bonds > shared pair

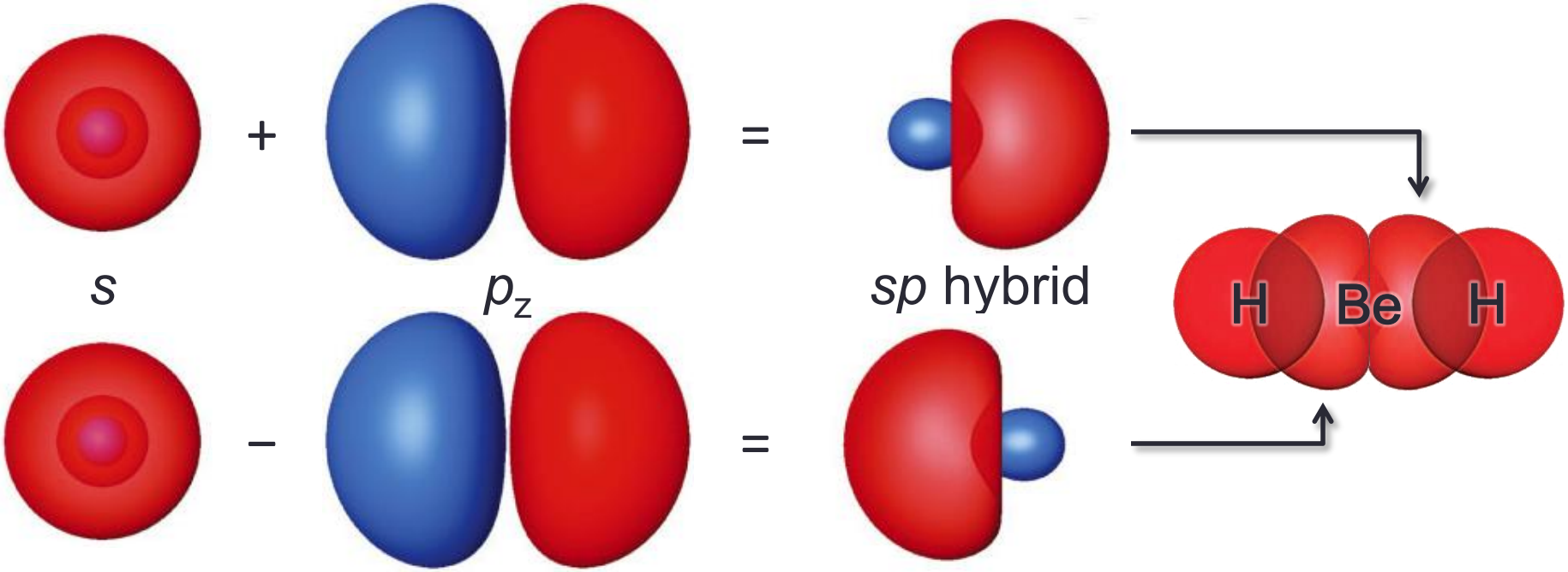
Example: SN = 2

H—Be—H BeH_2: $m = 2$, $n = 0$

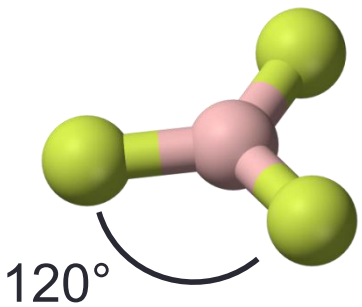
CO_2: $m = 2$, $n = 0$

:N≡N: N_2: $m = 1$, $n = 1$

Linear
直线形

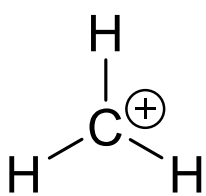


Example: SN = 3



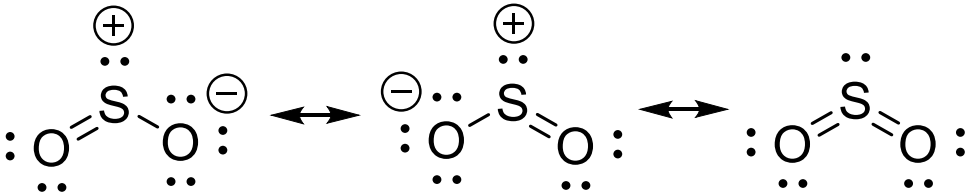
BF_3: $m = 3$, $n = 0$ Trigonal planar

平面三角形



CH_3^+: $m = 3$, $n = 0$ Trigonal planar

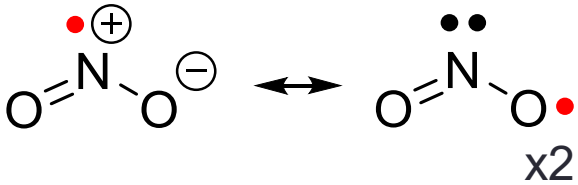
Example: SN = 3



SO_2: m = 2, n = 1

Bent / V-shaped

弯曲形 / V形 / 角形

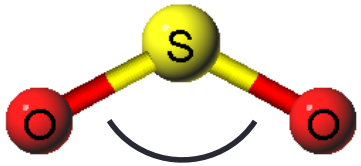


NO_2: m = 2, 0.5 < n < 1 Bent

134°

NO_2

VS



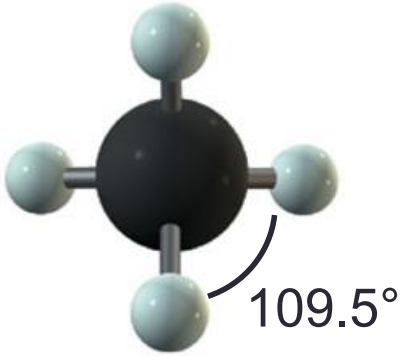
119°

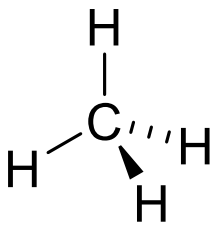
SO_2

117°

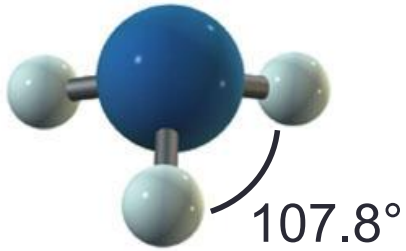
O_3

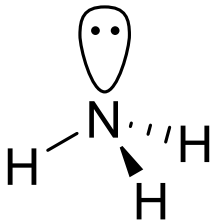
Example: SN = 4



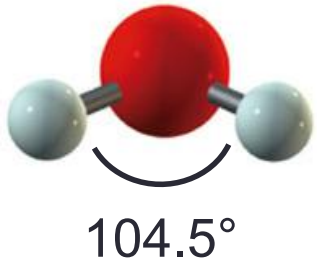


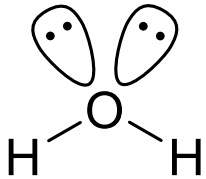
CH_4: $m = 4$, $n = 0$ Tetrahedral 四面体





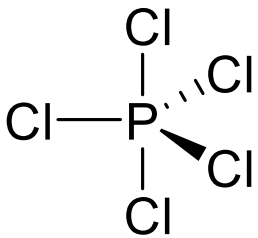
NH_3: $m = 3$, $n = 1$ Trigonal pyramidal 三角锥



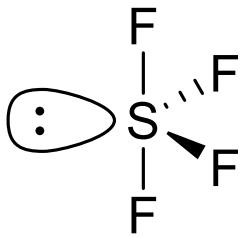


H_2O: $m = 2$, $n = 2$ Bent

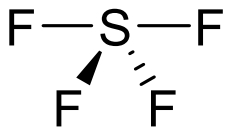
Example: SN = 5

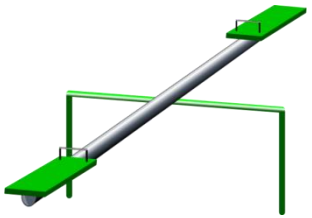


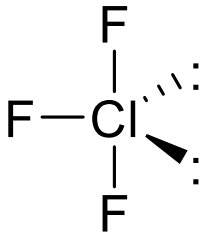
PCl_5: m = 5, n = 0 Trigonal bipyramidal 三角双锥



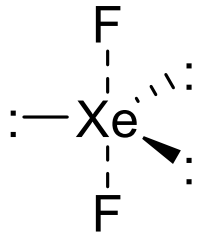
SF_4: m = 4, n = 1 Seesaw 跷跷板







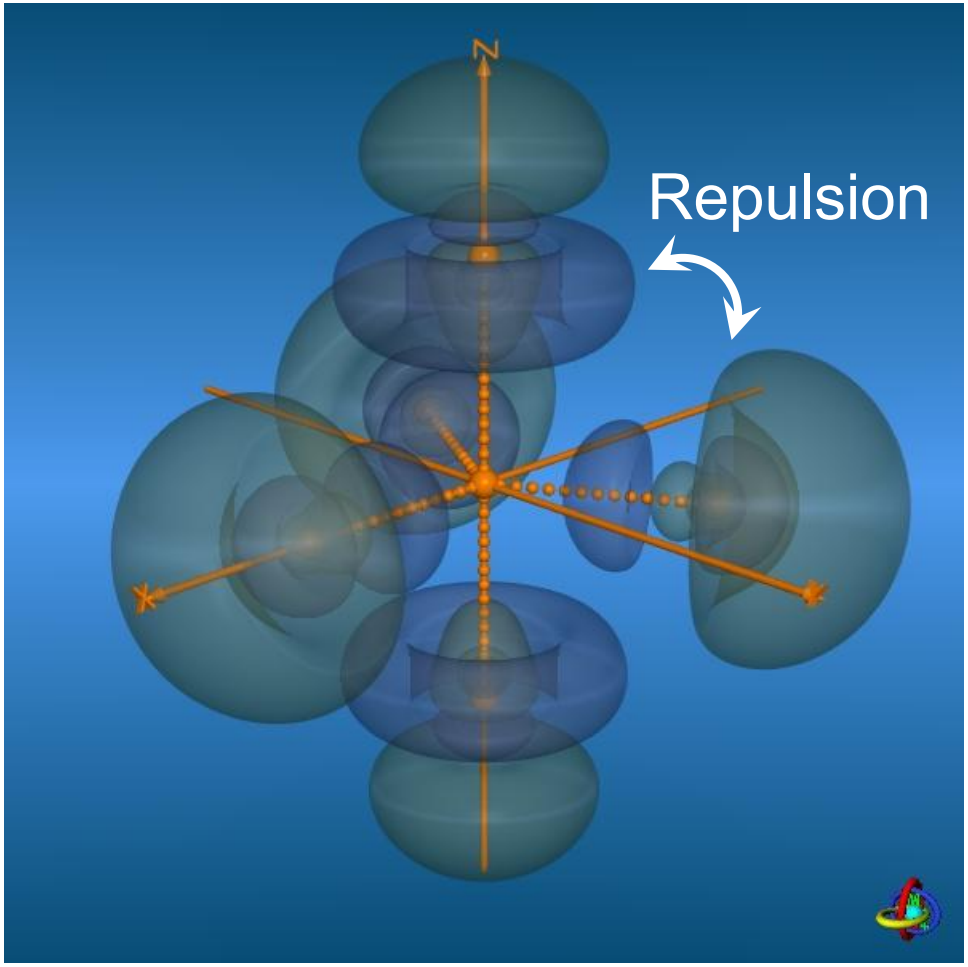
ClF_3: m = 3, n = 2 T-shaped



XeF_2: m = 2, n = 3 Linear

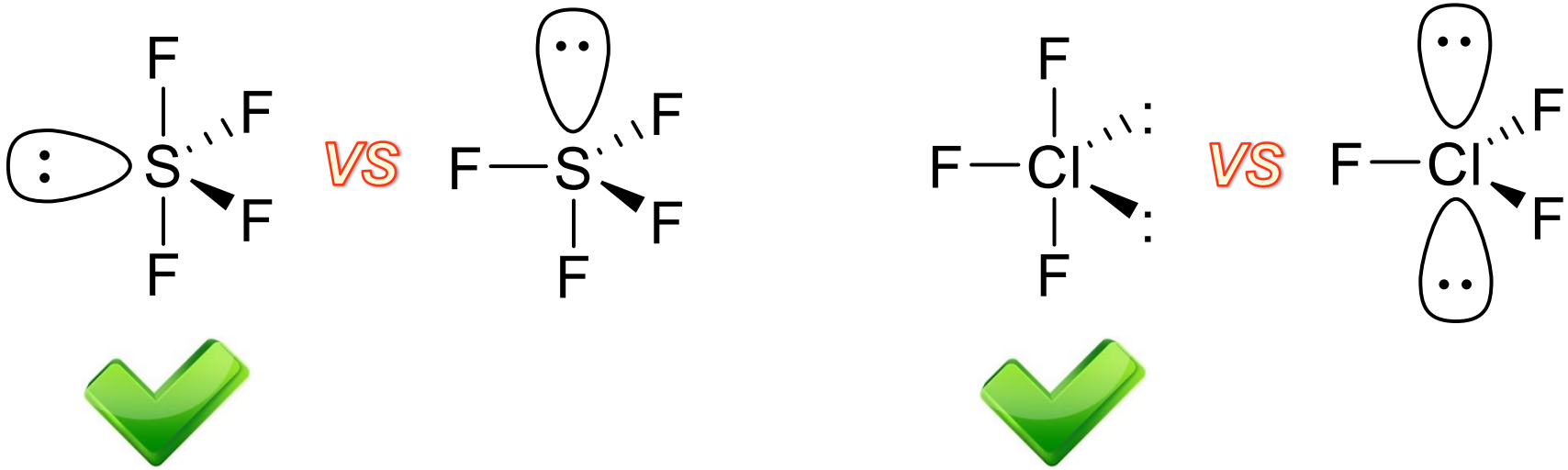
Example: sp^3d Hybridization

s, p_x, p_y, p_z, d_{z2}



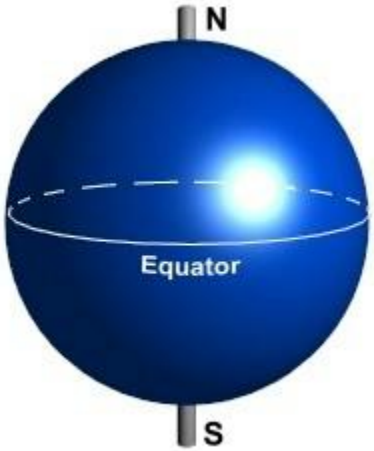
Avoid A–E ⊥ A–E
Minimize A–E ⊥ A–X

Lone Pairs Are Fat



Summary

1. Count the number of lone pairs from Lewis structure
2. Place the lone pairs on equatorial positions
3. Deduce the molecular geometry by ignoring the lone pairs



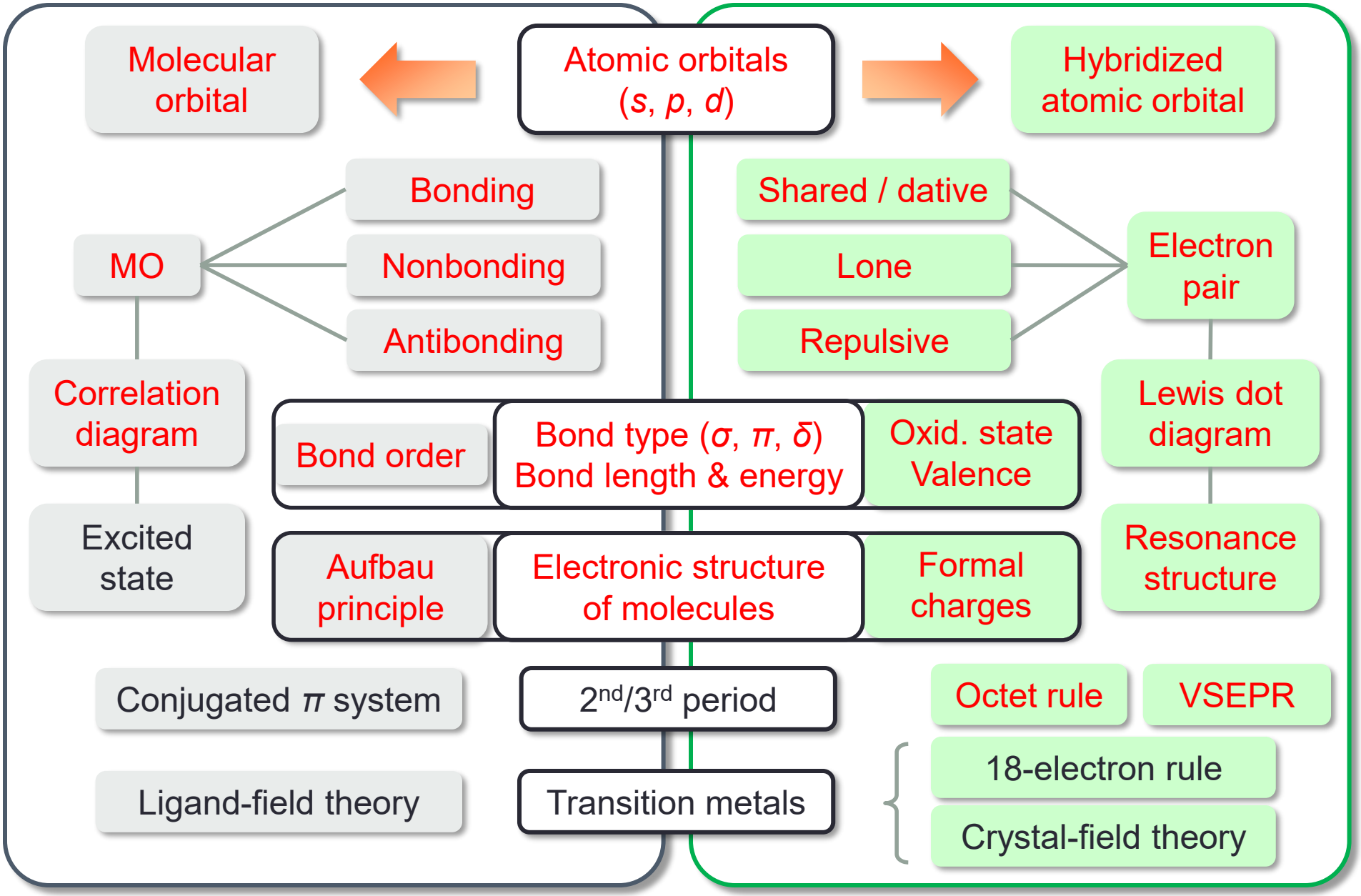
Simple and intuitive

Compatible with VB theory

Qualitative

Doesn't work for transition metals

Molecular Orbital (MO) | Valence Bond (VB) / Hybridization



Next: Organics

Reading: OGB8 §7